

PENETRATION TESTING REPORT

Enumeration & Network Scanning Phases

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Date	September 2026
Target / Client	networkwalks.com (written permission obtained) / Personal LAN Network
Permission Secured?	Yes — written authorization from Networkwalks
Environment	Kali Linux (Oracle VirtualBox) + Windows (Zenmap)
Phases Covered	Phase 1: Enumeration & Footprinting Phase 2: Network Scanning
Phases Pending	Phases 3–5 (Vulnerability Scanning, Exploitation, Post-Exploitation)

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1. Liability Disclaimer

All activities documented in this report were conducted exclusively on systems and networks for which prior written authorization was obtained. The enumeration activities targeting networkwalks.com were performed with explicit written permission from the domain owner. The network scanning activities were conducted solely on a personal LAN network comprising devices owned and operated by the tester.

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2. Introduction

This report covers two practical exercises representing the first two phases of a penetration testing engagement.

In the first exercise, passive and active enumeration was carried out against the public domain networkwalks.com using six Kali Linux tools: WHOIS, WhatWeb, Nslookup, Curl, Wafw00f, and DNSRecon. Each tool was used to extract a specific category of publicly accessible information, including domain registration data, web technology stack, DNS resolution, HTTP response headers, web application firewall presence, and DNS infrastructure records.

In the second exercise, active host discovery was performed on a personal LAN using Zenmap (the graphical interface for Nmap). A ping scan of the 192.168.1.0/24 subnet identified all live hosts and, where resolvable, their IP addresses, MAC addresses, and hardware vendor information.

Together, these exercises demonstrate the full transition from open-source intelligence gathering on a public target to active host enumeration on an authorized internal network — the foundational steps before any vulnerability assessment or exploitation phase.

3. Tools Used

Tool	Phase	Purpose
WHOIS	Enumeration	Domain registration info — registrar, creation/expiry dates, name servers, status flags
WhatWeb	Enumeration	Web technology fingerprinting — CMS, server, plugins, JS libraries, IP, email
Nslookup	Enumeration	DNS resolution of the target domain to its server IP address
Curl -I	Enumeration	HTTP response header inspection — server info, cookies, cache headers, API endpoints
Wafw00f	Enumeration	WAF detection — identifies whether a Web Application Firewall is protecting the site
DNSRecon	Enumeration	Full DNS record enumeration — SOA, NS, A, MX, TXT, SPF, SRV records
Zenmap (Nmap 7.991)	Network Scanning	Ping scan to discover live hosts, IP/MAC addresses, hostnames, and network topology

4. Activities Performed

4.1 Enumeration & Footprinting — networkwalks.com

The following enumeration activities were performed from inside Kali Linux running in Oracle VirtualBox. All six tools were run against networkwalks.com with written permission from Networkwalks. The objective of this phase was to collect as much publicly accessible information as possible before any active testing or exploitation.

4.1.1 WHOIS — Domain Registration Lookup

Command:

```
whois networkwalks.com
```

The WHOIS query returned the following registration data for NETWORKWALKS.COM:

- Registrar: GoDaddy.com, LLC | IANA ID: 146 | Abuse contact: abuse@godaddy.com / 480-624-2505
- Creation Date: 2019-11-06 | Last Updated: 2025-11-12 | Registry Expiry Date: 2027-11-06
- Registry Domain ID: 2452319255_DOMAIN_COM-VRSN
- Name Servers: NS6135.HOSTGATOR.COM, NS6136.HOSTGATOR.COM
- DNSSEC: unsigned (no cryptographic signing on DNS records)
- Domain Status: clientDeleteProhibited, clientRenewProhibited, clientTransferProhibited, clientUpdateProhibited

The registrar lock flags show the domain owner has applied protections against unauthorized transfer and modification. The absence of DNSSEC means DNS responses are not cryptographically signed, leaving the domain theoretically susceptible to DNS cache poisoning or spoofing attacks.

4.1.2 WhatWeb — Web Technology Fingerprinting

Command:

```
whatweb networkwalks.com
```

WhatWeb identified the following technologies across all redirect hops (HTTP 301 → HTTPS 200):

- Web Server: Apache (confirmed on all responses)
- CMS: WordPress 7.1
- Plugin: WordPress Download Manager 3.3.58
- JavaScript Libraries: jQuery 3.7.1, Bootstrap 7.1
- Resolved IP: 192.232.216.135 | Country: United States
- Email exposed in metadata: info@networkwalks.com
- Additional detected: Google-Tag-Manager, Open-Graph-Protocol, HTML5, Frame headers

The explicit plugin version (WP Download Manager 3.3.58) being publicly visible allows an attacker to query vulnerability databases such as WPScan or CVE Mitre for known exploits against that specific version without any further probing.

4.1.3 Nslookup — DNS Resolution

Command:

```
nslookup networkwalks.com
```

Results:

- DNS resolver used: 8.8.8.8 (Google Public DNS)
- Resolved IP address: 192.232.216.135 (non-authoritative answer)

The target resolves to a shared hosting IP (confirmed by the HostGator nameservers found in DNSRecon). Shared hosting means other customer websites likely reside on the same IP, which can have implications for cross-tenant information exposure in certain hosting misconfigurations.

4.1.4 Curl -I — HTTP Header Inspection

Command:

```
curl -I https://networkwalks.com
```

HTTP/2 200 response headers returned:

- server: Apache
- x-nginx-cache: WordPress
- content-type: text/html; charset=UTF-8
- referrer-policy: no-referrer-when-downgrade
- x-endurance-cache-level: 0
- date: Thu, 17 Sep 2026 08:21:12 GMT
- set-cookie: __wpdm_client (HttpOnly flag set — prevents JavaScript access to cookie)
- link: <https://networkwalks.com/wp-json/>; rel="https://api.w.org/" and <https://networkwalks.com/wp-json/wp/v2/pages/53>; rel="alternate" (WordPress REST API exposed)
- permissions-policy header present (private-state-token-redemption / issuance)

The Link header exposes the WordPress REST API endpoint /wp-json/. While this is default WordPress behavior, it enables unauthenticated enumeration of users, posts, and site metadata. The HttpOnly cookie flag is a positive security control, but the API exposure is an unnecessary information leak.

4.1.5 Wafw00f — WAF Detection

Command:

```
wafw00f networkwalks.com
```

Result:

- WAF Identified: ModSecurity (SpiderLabs)
- Requests made: 2

ModSecurity is a widely deployed open-source Web Application Firewall. Its presence indicates proactive traffic filtering is in place. However, identifying the specific WAF product and vendor allows an attacker to research ModSecurity-specific bypass techniques and tune their payloads accordingly before any exploitation attempt.

4.1.6 DNSRecon — DNS Infrastructure Enumeration

Command:

```
dnsrecon -d networkwalks.com
```

DNSRecon completed enumeration and returned 8 records:

- SOA: ns6135.hostgator.com — 50.87.144.87
- NS: ns6135.hostgator.com (50.87.144.87), ns6136.hostgator.com (192.232.216.131)
- Bind Version: "9.16.23-RH" (reported on both name servers)
- A Record: networkwalks.com → 192.232.216.135
- MX Record: mail.networkwalks.com → 192.232.216.135 (mail and web share same IP)
- TXT/SPF: v=spf1 +a +mx +ip4:50.87.144.87 +include:websitewelcome.com ~all
- TXT: google-site-verification token present
- SRV: Multiple _autodiscover._tcp entries → cpanelmaildiscovery.cpanel.net (184.94.x.x:443)

The hosting infrastructure is now fully identifiable as HostGator / cPanel. The SPF record uses ~all (softfail) rather than -all (hardfail), which means email spoofing attempts using this domain may still reach recipient inboxes under some mail server configurations. The mail server sharing the same IP as the web server is also notable — a compromise of one service could expose both.

4.2 Network Scanning — Personal LAN (192.168.1.0/24)

A ping scan was performed using Zenmap (Nmap 7.991) on Windows against the personal home LAN subnet. The goal was to enumerate all active hosts and identify their IP addresses, MAC addresses, and hardware vendors where identifiable.

Subnet scanned:

```
nmap -sn 192.168.1.0/24
```

Scan duration: 3.09 seconds | Total addresses probed: 256 | Live hosts found: 2

#	Hostname	IP Address	MAC Address	Vendor / Notes
1	Unit (Gateway)	192.168.1.1	2C:C1:F4:75:88:41	Nokia Solutions and Networks India Private Limited — LAN Gateway / Router
2	(Unnamed)	192.168.1.7	— (not returned)	No MAC/vendor resolved — host up, ownership to be verified

A network topology map was generated from the Zenmap Topology tab after the scan completed. The topology shows both discovered hosts (192.168.1.1 and 192.168.1.7) arranged around the central localhost scanning node, confirming a simple point-to-point/star topology on this subnet for the two hosts that responded to the ping sweep.

5. Risk Analysis

The table below summarizes all observations from both exercises and their potential security implications. These are findings from information gathering only — no exploitation was attempted. Presence of a version number or DNS record does not by itself confirm a vulnerability; further authorized testing would be required to confirm exploitability.

#	Finding	Tool	Potential Impact	Risk
1	WordPress plugin version publicly visible: WP Download Manager 3.3.58	WhatWeb	Attacker can query CVE databases for known exploits targeting that exact plugin version	Medium
2	WordPress REST API endpoint /wp-json/ exposed in HTTP Link header	Curl -I	Enables unauthenticated user enumeration, post/page metadata access via API	Medium
3	Full CMS and server stack fingerprinted (WordPress 7.1, Apache, jQuery 3.7.1, Bootstrap 7.1)	WhatWeb	Significantly reduces attacker reconnaissance time for targeted exploitation	Medium
4	DNSSEC not enabled on domain	WHOIS / DNSRecon	DNS responses not signed — domain susceptible to cache poisoning or spoofing in theory	Low
5	SPF record uses ~all (softfail) instead of -all (hardfail)	DNSRecon	Email spoofing using this domain may reach recipient inboxes in some mail server configurations	Medium
6	Hosting provider identifiable: HostGator / cPanel via NS and SRV records	DNSRecon	Allows attacker to target hosting-provider-specific vulnerabilities or social engineering	Low
7	WAF product identified: ModSecurity (SpiderLabs)	Wafw00f	Knowledge of specific WAF enables research of bypass techniques before exploitation	Low
8	Web and mail server share the same IP (192.232.216.135)	DNSRecon + Nslookup	Compromise of one service on the host may expose the other running on the same server	Low
9	Contact email exposed: info@networkwalks.com	WhatWeb	Can be used for phishing, social engineering, or targeted credential stuffing attacks	Low
10	2 live hosts discovered on personal LAN (gateway + one additional device)	Zenmap	Full device visibility — any unauthorized or unknown device should be investigated	Informational
11	Host at 192.168.1.1 identified as Nokia Solutions and Networks India Private Limited hardware ("Unit")	Zenmap	Vendor fingerprint confirms this host is the LAN gateway/router; should be verified against known equipment	Informational
12	Host at 192.168.1.7 returned no MAC address or vendor data	Zenmap	MAC/vendor could not be resolved locally (device outside ARP-resolvable range or filtered) — ownership should be confirmed manually	Informational

Risk Key: Medium (orange) | Low (gold) | Informational (grey)

6. Recommendations

#	Recommendation	Addresses
1	Suppress HTTP headers that expose CMS version, plugin names, and server software. Use a WordPress hardening plugin or server-level config (e.g. mod_headers) to strip or mask these values.	#1, #3
2	Restrict the WordPress REST API to authenticated users only. Disable unauthenticated access to /wp-json/users/ and other sensitive endpoints using a security plugin or custom code.	#2
3	Audit all installed WordPress plugins and keep them updated. Subscribe to the WPScan vulnerability database for notifications when new CVEs are published for installed plugins.	#1
4	Enable DNSSEC on networkwalks.com through the GoDaddy registrar control panel to cryptographically sign DNS records and protect against cache poisoning.	#4
5	Update the SPF TXT record from ~all (softfail) to -all (hardfail) to reject unauthorized senders and reduce the risk of email spoofing using this domain.	#5
6	Consider moving the mail server to a separate IP address to isolate it from the web server, limiting the blast radius if either service is compromised.	#8
7	Perform regular internal LAN scans to maintain an up-to-date device inventory. Investigate any unrecognized or unauthorized devices promptly.	#10, #11, #12
8	Enable network access controls on the router (e.g. MAC address filtering or guest network isolation) to limit which devices can communicate on the main LAN.	#10, #12

7. Conclusion

This report documented two foundational phases of a penetration testing methodology: enumeration and footprinting of a public web target, and active host discovery on a personal LAN network.

During the enumeration phase, six Kali Linux tools were used to gather publicly available information about networkwalks.com. The exercise confirmed that even before any active exploitation, a significant amount of infrastructure detail is freely accessible — including the CMS and plugin versions, hosting provider, name server software, mail server configuration, WAF implementation, and administrative contact information.

During the network scanning phase, Zenmap successfully identified 2 live hosts on the 192.168.1.0/24 subnet — the LAN gateway (192.168.1.1, fingerprinted as Nokia Solutions and Networks India Private Limited hardware) and one additional host (192.168.1.7) for which MAC/vendor data could not be resolved. A topology map was generated to visualize the network layout.

Key takeaways:

- Information gathering is one of the most critical pre-exploitation phases. The data collected in these two exercises directly informs the approach for all subsequent penetration testing phases.
- Public-facing infrastructure exposes considerably more information than most administrators realize, often without requiring any active probing or exploitation.
- Even a small-scale LAN ping sweep quickly produces a useful picture of the internal network — highlighting why device inventory management and network access controls are essential, regardless of how many hosts are online at scan time.
- All findings in this report are observations from authorized enumeration only. No exploitation was attempted at any stage. Confirmation of actual vulnerabilities would require further authorized penetration testing.

Phases 3 through 5 (vulnerability scanning, exploitation, and post-exploitation) are currently in progress and will be documented in separate reports.

8. Evidence / Screenshots

The screenshots below were captured during the practical exercises and serve as primary evidence for all findings described in Section 4.

File	Tool	What it shows
Screenshot_2026-09-17_134109.png	WHOIS	Full WHOIS output for networkwalks.com — registrar, dates, name servers, domain status flags
Screenshot_2026-09-17_134637.png	WhatWeb	WhatWeb fingerprint output across all redirects — CMS, plugins, server, jQuery, email
Screenshot_2026-09-17_135030.png	Nslookup	DNS resolution of networkwalks.com resolving to 192.232.216.135
Screenshot_2026-09-17_135137.png	Curl -I	HTTP/2 200 response headers — server, cache, cookie, Link header with /wp-json/
Screenshot_2026-09-17_135305.png	Wafw00f	WAF detection confirming ModSecurity (SpiderLabs) is protecting the site
Screenshot_2026-09-17_135446.png	DNSRecon	Full DNS enumeration — 8 records returned including SOA, NS, A, MX, TXT, SPF, SRV
Screenshot_2026-09-26_192952.png	Zenmap	Ping scan output listing the 2 live hosts with IPs and MAC/vendor data
Screenshot_2026-09-26_193653.png	Zenmap Topology	Network topology diagram from the Zenmap Topology tab showing both discovered hosts around the scanning node

Evidence — Enumeration & Footprinting (networkwalks.com)

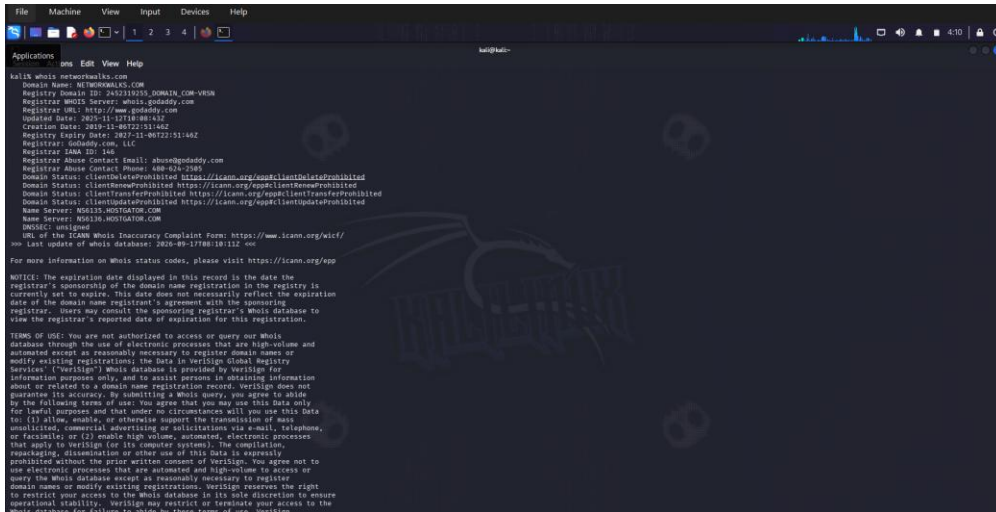


Fig. 1 — WHOIS output for networkwalks.com

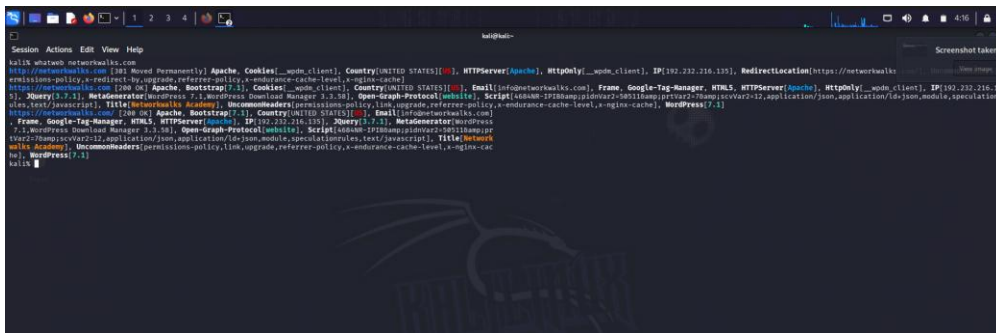


Fig. 2 — WhatWeb fingerprint output

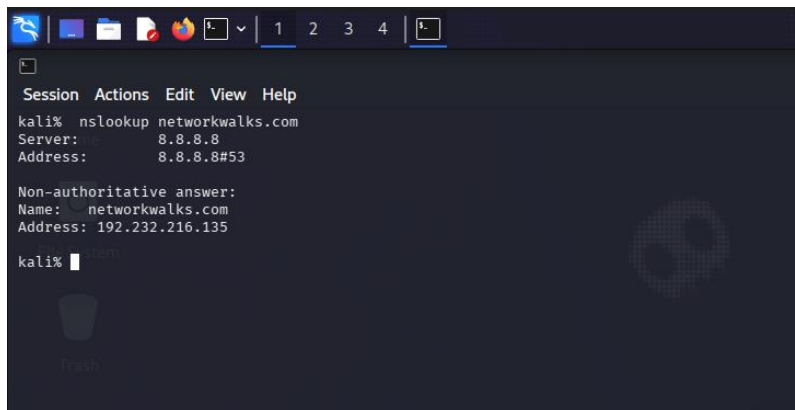


Fig. 3 — *Nslookup* DNS resolution

```
kali@kali~$ curl -I https://networkwalks.com
HTTP/2 200
permissions-policy: private-state-token-redemption=(self "https://www.google.com" "https://www.gstatic.com" "https://recaptcha.net" "https://challenges.cloudflare.com" "https://hcaptcha.com"), private-state-token-issuance=(self "https://www.google.com" "https://www.gstatic.com" "https://recaptcha.net" "https://challenges.cloudflare.com" "https://hcaptcha.com")
link: <https://networkwalks.com/wp-json/?rel=swagger>, <https://networkwalks.com/wp-json/wp/v2/pages/5>; rel="alternate", title="200", type="application/json", <https://networkwalks.com/?rel=shortlink>
set-cookie: __wsm_client=3fcaad959d9c8b8a29f358dffa; path=/; domain=networkwalks.com; secure; HttpOnly
referrer-policy: no-referrer
x-endurance-cache-level: 8
x-endurance-cache: 00000000000000000000000000000000
content-type: text/html; charset=UTF-8
Date: Tue, 23 Sep 2020 08:21:12 GMT
server: Apache
```

Fig. 4 — Curl -I HTTP response headers

```
kali@kali~$ wafw00f networkwalks.com
[+] Checking https://networkwalks.com
[+] The site https://networkwalks.com is behind ModSecurity (SpiderLabs) WAF.
[-] Number of requests: 2
```

Fig. 5 — Wafw00f WAF detection (ModSecurity)

```
kali@kali~$ dnsrecon -d networkwalks.com
2026-09-17T04:23:51.935201-0400 INFO std: Starting enumeration for domain: networkwalks.com
2026-09-17T04:23:51.936027-0400 INFO std: Performing General Enumeration against: networkwalks.com ...
2026-09-17T04:23:53.074801-0400 ERROR No answer for DNSSEC query for networkwalks.com
2026-09-17T04:23:53.715177-0400 INFO SOA ns6135.hostgator.com 50.87.144.87
2026-09-17T04:23:54.448479-0400 INFO NS ns6136.hostgator.com 192.232.216.131
2026-09-17T04:23:55.136501-0400 INFO Bind Version for 192.232.216.131 "9.16.23-RH"
2026-09-17T04:23:55.136843-0400 INFO NS ns6135.hostgator.com 50.87.144.87
2026-09-17T04:23:55.736691-0400 INFO Bind Version for 50.87.144.87 "9.16.23-RH"
2026-09-17T04:23:56.631040-0400 INFO MX mail.networkwalks.com 192.232.216.135
2026-09-17T04:23:57.121089-0400 INFO A networkwalks.com 192.232.216.135
2026-09-17T04:23:58.294464-0400 INFO TXT networkwalks.com google-site-verification=rr4eRmqHoWY3XemniZDNV4q75X-Ij-mjgEg-UsYI
2026-09-17T04:23:58.294737-0400 INFO TXT networkwalks.com vnsplf1 +a +mx +ip4:50.87.144.87 +include:websitewelcome.com ~all
2026-09-17T04:23:59.018884-0400 INFO Enumerating SRV Records
2026-09-17T04:24:03.008145-0400 INFO SRV _autodiscover._tcp.networkwalks.com cpanelmaildiscovery.cpanel.net 184.94.204.9 443
2026-09-17T04:24:03.008526-0400 INFO SRV _autodiscover._tcp.networkwalks.com cpanelmaildiscovery.cpanel.net 184.94.203.15 443
2026-09-17T04:24:03.008608-0400 INFO SRV _autodiscover._tcp.networkwalks.com cpanelmaildiscovery.cpanel.net 184.94.204.11 443
2026-09-17T04:24:03.008717-0400 INFO SRV _autodiscover._tcp.networkwalks.com cpanelmaildiscovery.cpanel.net 184.94.204.15 443
2026-09-17T04:24:03.008937-0400 INFO SRV _autodiscover._tcp.networkwalks.com cpanelmaildiscovery.cpanel.net 184.94.203.9 443
2026-09-17T04:24:03.009169-0400 INFO SRV _autodiscover._tcp.networkwalks.com cpanelmaildiscovery.cpanel.net 184.94.204.14 442
2026-09-17T04:24:03.009262-0400 INFO SRV _autodiscover._tcp.networkwalks.com cpanelmaildiscovery.cpanel.net 184.94.203.11 443
2026-09-17T04:24:03.009426-0400 INFO SRV _autodiscover._tcp.networkwalks.com cpanelmaildiscovery.cpanel.net 184.94.203.14 443
2026-09-17T04:24:03.010710-0400 INFO 8 Records Found
2026-09-17T04:24:03.011055-0400 INFO Completed enumeration for domain: networkwalks.com
```

Fig. 6 — DNSRecon full DNS enumeration

Evidence — Network Scanning (Personal LAN)

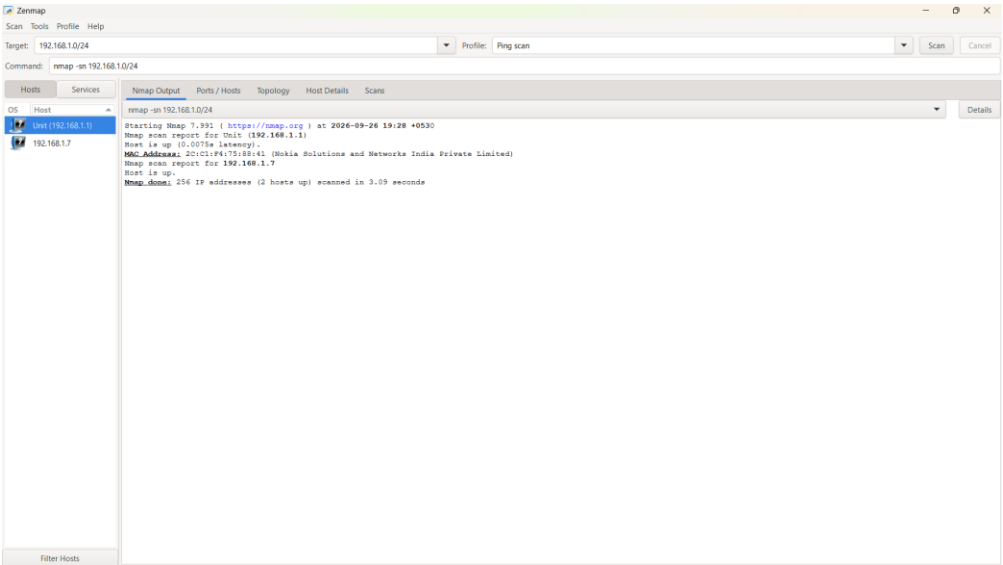


Fig. 7 — Zenmap ping scan output (2 hosts up)

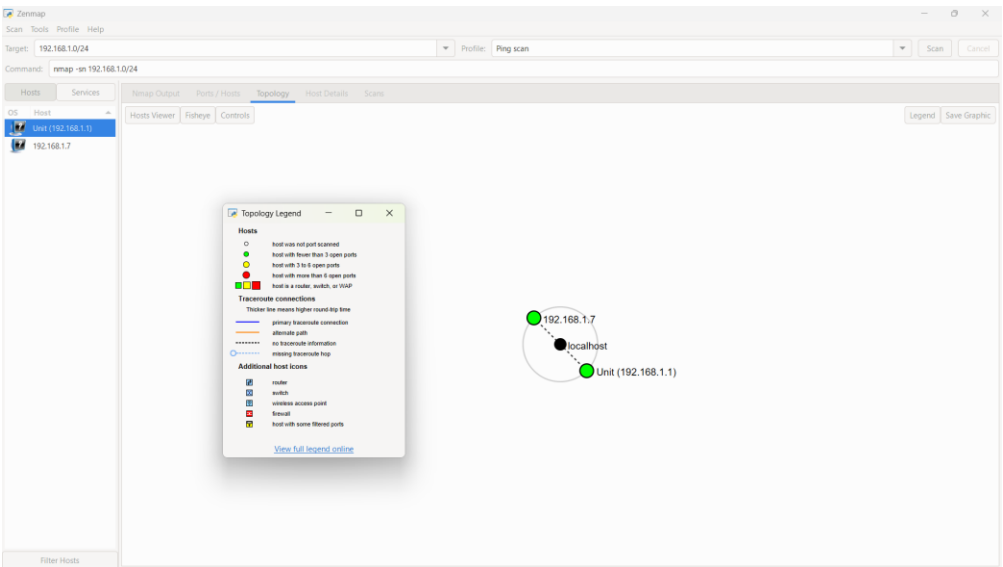


Fig. 8 — Zenmap topology map of discovered hosts

— End of Report —

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